

Predicting Student Success: A Data Mining Approach to Retention in Higher Education

EXECUTIVE SUMMARY

Student attrition represents a systemic inefficiency in higher education, resulting in lost potential for learners and significant resource drain for institutions. This report details a Data Mining project aimed at transitioning retention strategies from reactive observation to proactive intervention. By analyzing a dataset of 4,424 undergraduates, this study identifies critical "tipping points" in academic momentum and financial stability. Utilizing a calibrated Decision Tree classifier, we achieved a ROC AUC of 0.835, demonstrating a strong capability to rank at-risk students. The findings suggest that early academic performance (1st Semester) and tuition payment status are the strongest predictors of dropout, overriding pre-admission metrics like high school grades.

1. INTRODUCTION

1.1 The Problem Space

Higher education institutions traditionally rely on reactive retention models— intervening only after a student has failed a semester or formally withdrawn. This approach is often too late to alter the student's trajectory. To improve graduation rates, administrators require a **predictive framework** capable of identifying at-risk students while they are still enrolled.

1.2 Research Questions

This analysis is guided by four specific research questions (RQs) designed to diagnose the root causes of attrition and test the feasibility of automated detection:

- **RQ1 (Academic Diagnostic):** Is academic performance in the first semester— specifically the number of approved curricular units—a reliable early warning signal for future attrition?
- **RQ2 (Socio-Economic Diagnostic):** To what extent do financial factors, specifically tuition payment status and scholarship holding, correlate with student success rates?
- **RQ3 (Demographic Diagnostic):** Do pre-existing factors such as age enrollment and high school admission grades significantly differentiate dropouts from graduates?
- **RQ4 (Predictive Modeling):** Can a machine learning model accurately predict atrisk students in real-time, despite data imbalance challenges?

1.3 Relevance

The relevance of this study lies in **resource optimization**. By validating these questions, university administrators can move away from generic support emails and toward targeted, data-driven interventions for the students who need them most.

2. DATA DESCRIPTION

2.1 Source and Scope

The analysis utilizes the **Student Academic Performance Dataset** sourced from the UCI Machine Learning Repository. It contains **4,424 records** of undergraduate students enrolled in various degree programs (e.g., Agronomy, Design, Management).

2.2 Key Variables

The dataset consists of **36 features** categorized into three pillars:

1. **Academic Performance (Curricular Units):** This section tracks the student's workload and success. Key variables include:
 - *Enrolled*: Number of courses attempted.
 - *Evaluations*: Number of exams taken.
 - *Approved*: Number of courses passed (The critical metric).
 - *Grade*: The average GPA (0-20 scale).
2. **Socio-Economic Status:** Indicators of financial health, including *Tuition fees up to date* (Binary), *Scholarship holder* (Binary), and macroeconomic indicators like *Unemployment Rate* and *GDP*.
3. **Demographics:** Personal background data including *Age at enrollment*, *Gender*, *Marital Status*, and *Displaced* status (living away from home).

2.3 Data Processing

Prior to analysis, the data underwent rigorous preprocessing to ensure quality:

- **Cleaning:** Column names were standardized to remove whitespace and special characters.
- **Feature Engineering:** A "Success Rate" variable was derived for the 1st semester ($\text{Approved Units} / \text{Enrolled Units}$) to normalize performance across different major workloads.
- **Encoding:** Categorical variables were transformed using Label Encoding.
- **Scaling:** Numerical features were standardized using a StandardScaler to prevent features with larger ranges (e.g., Admission Grade) from dominating the model.

3. ANALYSIS OF DATA

3.1 Methodology

The analytical approach was divided into two phases: **Diagnostic Exploratory Data Analysis (EDA)** and **Predictive Modeling**.

Challenges Tackled:

The primary challenge was Class Imbalance. The dataset contained significantly more "Graduates" than "Dropouts." Training a model on raw data would result in a bias toward the majority class (high accuracy, but zero recall for dropouts). **Data Mining Methods Used:**

1. **Hybrid Resampling (SMOTE-ENN):** We applied *Synthetic Minority Oversampling Technique* (SMOTE) combined with *Edited Nearest Neighbors* (ENN). This generated synthetic examples of dropouts to balance the training data while cleaning noisy decision boundaries.
2. **Algorithm Selection:** A **Decision Tree Classifier** was chosen for its interpretability ("White Box" nature), allowing stakeholders to trace the logic behind a risk flag.
3. **Hyperparameter Optimization:** We utilized **RandomizedSearchCV** to explore thousands of tree configurations (Depth, Split Criteria, Leaf Nodes). This method outperformed standard Grid Search in our tests.
4. **Probability Calibration:** The final model was calibrated using **Sigmoid (Platt) Scaling** to ensure that risk scores (probabilities) were statistically reliable.

4. RESULTS

4.1 Diagnostic Findings (EDA)

RQ1: The Academic Tipping Point

Analysis reveals a stark separation in first-semester performance. As shown in Figure 1, the median number of approved units for Dropouts is near zero, whereas Graduates consistently approve the majority of their enrolled units.

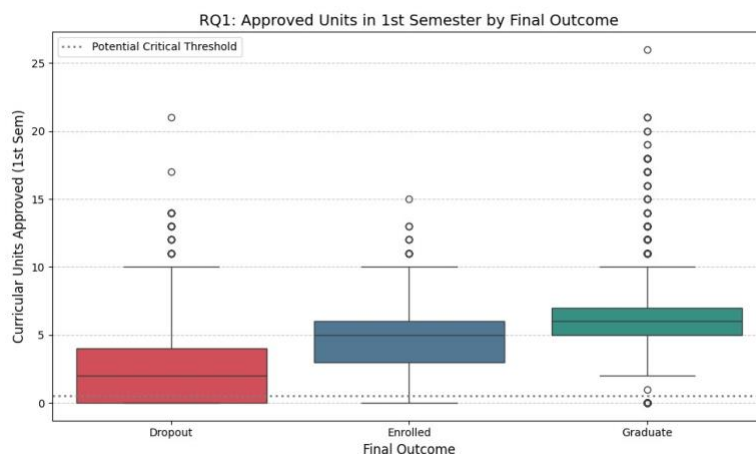


Figure 1: Distribution of Approved Curricular Units in 1st Semester by Final Outcome.

Finding: Students who approve fewer than 50% of their enrolled units in the first semester are at immediate, critical risk.

RQ2: The Financial Barrier

Financial stability acts as a prerequisite for retention. Figure 2 demonstrates that students with "Late/Missing" tuition payments have a success rate below 40%. Conversely, scholarship holders show consistently higher success rates, suggesting that financial aid acts as a protective buffer against attrition.

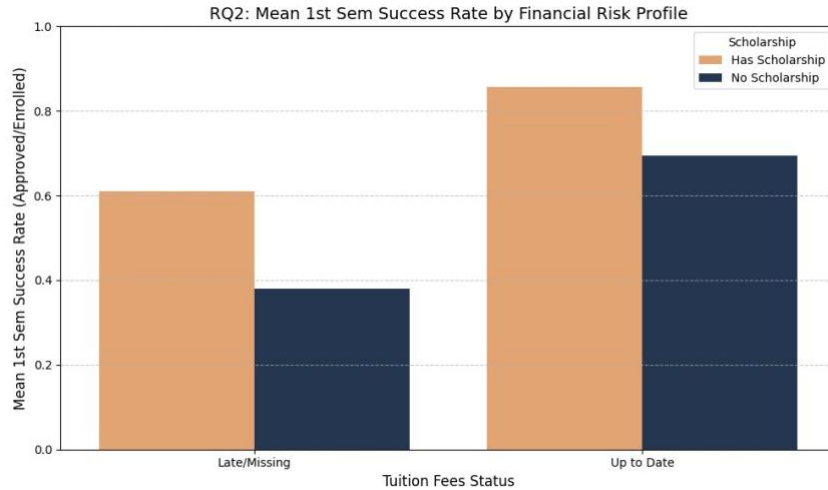


Figure 2: Mean 1st Semester Success Rate grouped by Financial Risk Profile.

RQ3: Demographics vs. History

A comparison of "Age" vs. "Admission Grade" (Figure 3) reveals that Dropouts tend to be older (delayed enrollment). However, their Admission Grades (High School GPA) are nearly identical to Graduates.

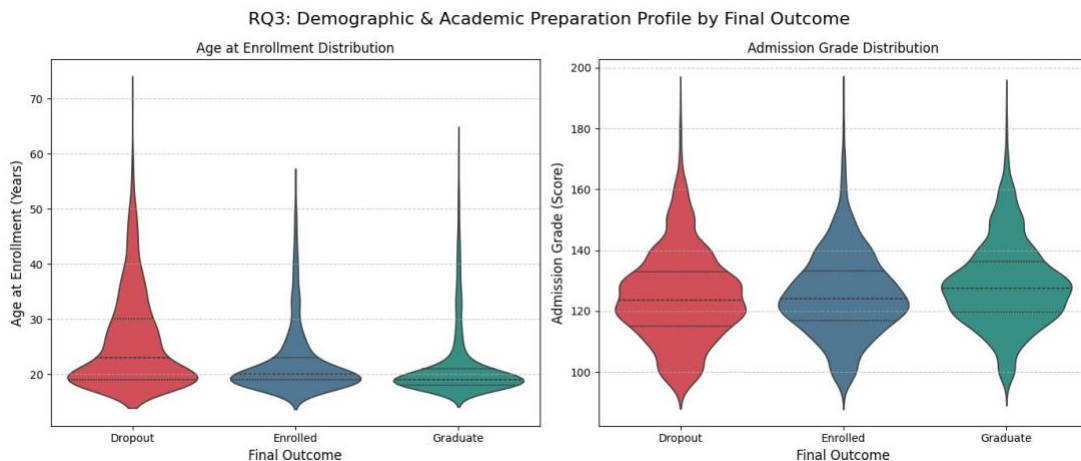


Figure 3: Demographic and Academic Preparation Profiles (Violin Plots).

Finding: A student's high school history is **not** a reliable predictor of university retention. Current life circumstances (Age) and current performance matter more.

4.2 Predictive Modeling Results (RQ4)

To validate RQ4, the Decision Tree model was optimized and tested on unseen data.

Model Performance:

- **ROC AUC Score: 0.835**
- **Final Accuracy: 62.37%** (Trade-off for high Recall)
- **F1 Score: 60.67%**

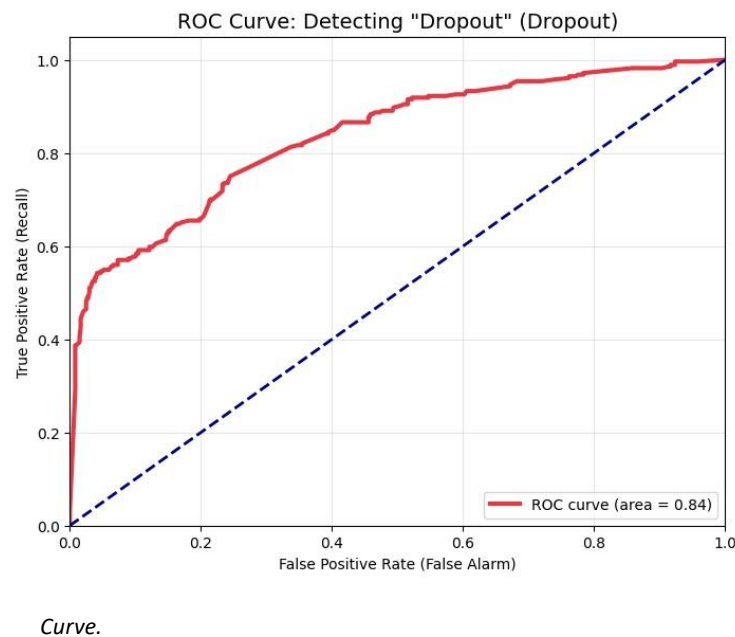


Figure 4: Receiver Operating Characteristic (ROC)

Interpretation:

The ROC AUC of 0.835 indicates a strong discriminative ability. There is an 83.5% probability that the model successfully assigns a higher risk score to a random dropout than to a random graduate.

Strategic Thresholding:

To prioritize the identification of at-risk students, we adjusted the decision threshold from the standard 0.50 to 0.471. This intentional bias lowered overall accuracy but maximized Recall (Sensitivity). In a retention context, a False Positive (counseling a safe student) is acceptable, but a False Negative (missing a dropout) is a failure.

Feature Importance:

The model identified the following variables as the strongest drivers of retention (Figure 5):

1. **Curricular Units 2nd Sem (Grade)**
2. **Curricular Units 1st Sem (Approved)**
3. **Age at Enrollment**

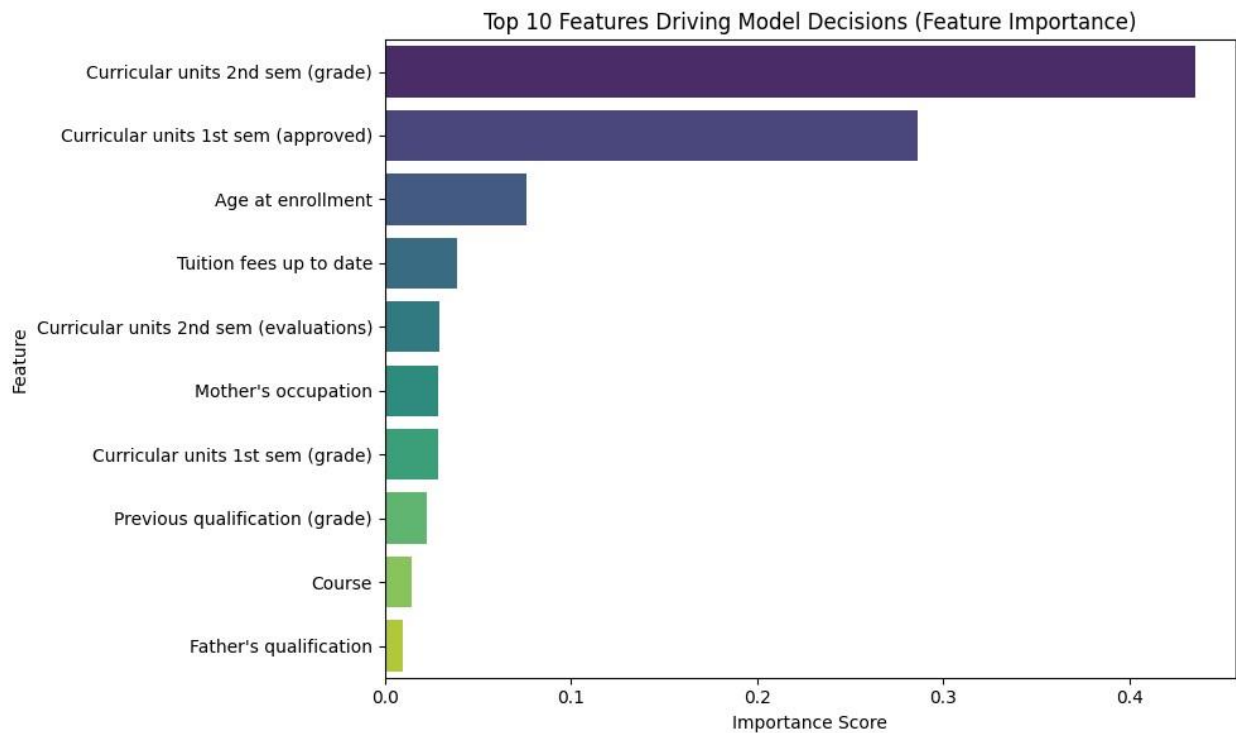


Figure 5: Top 10 Features Driving Model Decisions.

5. CONCLUSIONS

This study confirms that student attrition is not random; it is a predictable phenomenon driven by identifiable academic and financial patterns.

Key Conclusions:

1. **The "Tipping Point" is Early:** The transition from the 1st to the 2nd semester is critical. Academic momentum in the first year is the single strongest predictor of graduation.
2. **Financial Health is a Proxy for Retention:** Unpaid tuition is an immediate red flag that often precedes academic disengagement.
3. **Prediction is Viable:** Using a calibrated machine learning model, we can rank students by risk with high reliability (AUC 0.835), allowing for targeted rather than generic intervention.

5.1 Recommendations

Based on the numerical evidence, the following actions are recommended:

- **Implement Financial Triggers:** Automated counseling outreach should occur immediately when a student misses a tuition deadline.
- **Academic Support Blocks:** Mandatory tutoring interventions should be assigned to any student approving fewer than 50% of their units in Semester 1.
- **Algorithmic Risk Scoring:** Deploy the predictive model to generate weekly "Risk Lists" for academic advisors, prioritizing outreach based on the model's probability scores.

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